Chapter 1: Introduction

We will discuss, in this chapter, some basic concepts for experimental design such as:

- Response
- Factor
- Replicates
- Randomization
- Blocking

Definition: An **experiment** is a scientific procedure undertaken to make a discovery or to determine something.

Example 1: A design engineer is interested in determining any difference in the mean of burning times of chemical flares of two different formulations. She found the burning times of 10 flares for each of the two formulations. The results are given in the following table:

Type1	Type2
65	64
82	56
81	71
67	69
57	83
59	74
66	59
75	82
82	65
70	79

Factor with two levels

Response Variable

The measurable outcome of the experiment is called **response**.

The variables whose effect on the outcome is to be studied is called **factor**.

This is a **single factor** experiment with 10 **replicates**.

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The variables whose effect on the outcome is to be studied is called **factor**.

In order to study the effect of a factor on the response, two or more values of the factor are used. These values are referred to as **levels** of the factor.

Many experiments involve the study of the effect of two or more factors. A **treatment** is combinations of factor levels.

Example 2: Suppose we are interested in examining the effect of two levels of nitrogen at 40 and 60 pounds per plot (N40, N60) and two levels of phosphorus at 10 and 20 pounds per plot (P10 and P20) on the yield of a crop. The four level factor-level combinations (treatments) that are assigned **at random** to the experimental units are:

1. N40-P10 2. N40-P20 3. N60-P10 4. N60-P20

Example 3: Suppose two keyboards denoted by A and B are being compared in terms of typing efficiency. Six different manuscripts denoted by 1-6 are given to the same typist. Suppose the test is arranged in the following sequence:

1. A, B 2. A, B 3. A, B 4. A, B 5. A, B 6. A, B

After typing the manuscript on keyboard A, the typist will be familar with the content of the manuscript when he or she is typing the same manuscript on keyboard B.

It is **impossible** to disentangle the learning effect from treatment effect.

Randomization could help reduce the unwanted influence of learning effect, which might have not be known to the investigator who planned the study. By randomization the typing order for each manuscript, the test sequence may appear as follow:

1. A, B 2. B, A 3. A, B 4. B, A 5. A, B 6. A, B

Example 3 (Cont.): By randomization the typing order for each manuscript, the test sequence may appear as follow:

1. A, B 2. B, A 3. A, B 4. B, A 5. A, B 6. A, B

The design is **not balanced** because B benefits from the learning effect in four trials while A only benefits from two trials.

The leaning effect can be completely eliminated by requiring that half of trials have the order A, B and the other half the order B, A. The actual assignment of A, B and B, A to the six manuscripts should be done by randomization. This method is referred to to a **balanced randomization**.

By balanced randomization the typing order for each manuscript, the test sequence may appear as follow:

1. A, B 2. A, B 3. B, A 4. B, A 5. A, B 6. B, A

Example 3 (Cont.): By balanced randomization the typing order for each manuscript, the test sequence may appear as follow:

1. A, B 2. A, B 3. B, A 4. B, A 5. A, B 6. B, A

Randomization is the corenerstone underlying the use of statistical methods in experimental design.

Both the **allocation** of the experimental material and the **order** in which the individual runs or trials of the experiment are to be performed are randomly determined.

By properly randomizing the experiment, we assist in **averaging out** the effects of extraneous factors that may be present.

Another purpose of randomization is to **eliminate** bias.

Statistical methods require that observations (or errors) be independently distributed variables. Randomization usually makes this assumption valid.

Example 3 (Cont.): By balanced randomization the typing order for each manuscript, the test sequence may appear as follow:

1. A, B 2. A, B 3. B, A 4. B, A 5. A, B 6. B, A

Because the manuscripts can vary in length and difficulty, each manuscript is treated as a **block** with two keyboards as two tratments.

The experiment is replicated six times (with six manuscripts) and blocking is used to compare the two keyboards with the same manuscripts.

Blocking is a technique used to increase the precision of an experiment. A block is a portion of the experimental material that should be more **homogeneous** than the entire set of materials.

Blocking involves making comparisons among the condition of interest in the experiment within each block.

Example 3 (Cont.): By balanced randomization the typing order for each manuscript, the test sequence may appear as follow:

1. A, B 2. A, B 3. B, A 4. B, A 5. A, B 6. B, A

For simplicity of discussion, we have assumed that only one typist was involved in the experiment. In a pratical situation, such an experiment should involve **several typist** that are representative of the population of typists so that the conclusion made from the study would apply more generally.

The **three basich principles** of experimental design are:

- **Replication**
- **Randomization**
- **Blocking**

Definition: An **experiment** is a scientific procedure undertaken to make a discovery or to determine something.

The general approach to planning and conducting the experiment is called the **strategy of experimentation**.

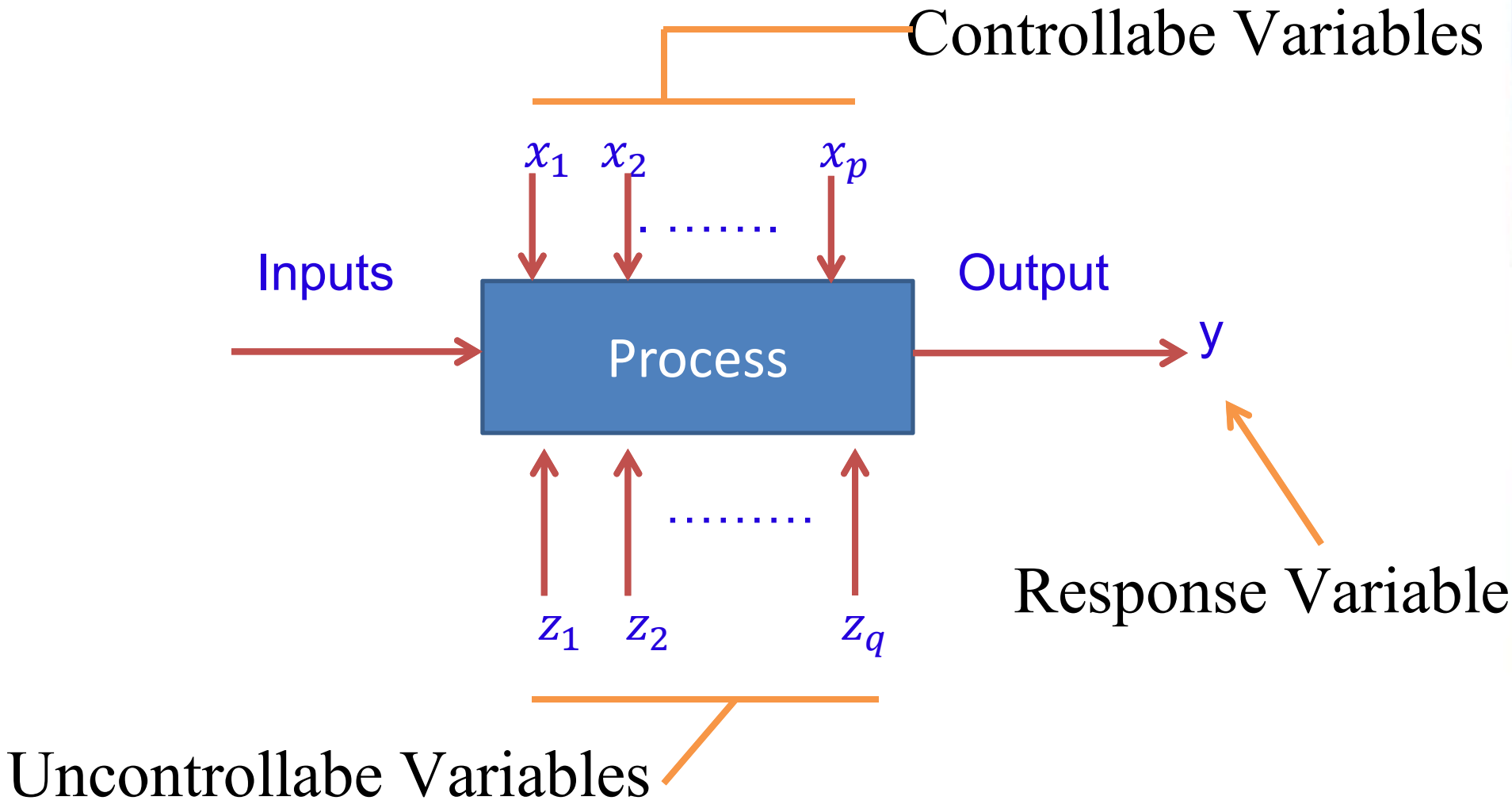
The **design of experiments** (**DOE**) is an efficient procedure for planning experiments so that the data obtained can be analyzed to yield valid and objective conclusions.

The **statistical design of experiment** is the process of planning the experiment so that the appropriate data that can be analyzed by statistical methods will be collected.

When you are doing "statistics" do **not** forget:

- what you and your colleagues know about the subject matter field. Statistical techniques are useless unless combined with appropriate subject matter knowledge and experience. They are an adjunct to, not a replacement for, subject matter expertise.
- keep the design as simple as possible.

Definition: An **experiment** is a scientific procedure undertaken to make a discovery or to determine something.



Although an experiment is repeated under as nearly as possible identical conditions, the observed results are not identical.

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Experimental problems can be classified into five broad categories according to their objectives.

1. **Treatment Comparisons:** The main purpose is to compare several treatments and select the best.
2. **Variable Screening:** If there is a large number of variables in a system but only a relatively small number of them is important, a screening experiment can be conducted to identify important variables.
3. **Response surface Exploration:** The relationship between the response and important selected variables is sometimes referred to as a response surface.
4. **System optimization:** In many investigation, interest lies in the optimization of the system.
5. **System Robustness:** Besides optimizing the response, it is important in quality improvement to make the system robust against noise variation or uncontrivable variables.

Definition: An **experiment** is a scientific procedure undertaken to make a discovery or to determine something.

Experiment that we are going to study:

- Those with **one** (not multiple one) **measurable** (**quantitative**) outcome.
- Those involving uncertainty
- Those involving some comparison (comparative experiments)

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The fluctuation that occurs from one repetition to another is called **noise**, **experimental variation**, **experimental error**, or merely **error**.

We aim to cope with uncertainty using statistical methods.

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Statistical Model: $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ where $i = 1,2$ and $j = 1,2,\ldots,10$

Response Variable

Overall Mean

Treatment Effect

Random Error

Null Hypothesis: $H_0: \tau_i = 0$ for $i = 1,2$ (There is **no** treatment effect.)

Alternative Hypothesis:

- $H_A: \tau_i \neq 0$ for $i = 1,2$ (There **is** a treatment effect.)
- $H_A: \tau_i > 0$ for $i = 1,2$ (There is a **positive** treatment effect.)
- $H_A: \tau_i < 0$ for $i = 1,2$ (There is a **negative** treatment effect.)

Statistical Model: $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ where $i = 1,2$ and $j = 1,2,\ldots,10$

Let τ_1+τ_2=0→$\tau_2 = -\tau_1$

$$Y = \begin{bmatrix} y_{11} \\ y_{12} \\ . \\ . \\ . \\ y_{110} \\ y_{21} \\ y_{22} \\ . \\ . \\ . \\ y_{210} \end{bmatrix}, X = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ . & . \\ . & . \\ . & . \\ 1 & 1 \\ 1 & -1 \\ 1 & -1 \\ . & . \\ . & . \\ . & . \\ 1 & -1 \end{bmatrix}, \beta = \begin{bmatrix} \mu \\ \tau_1 \end{bmatrix}, E = \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ . \\ . \\ . \\ \varepsilon_{110} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ . \\ . \\ . \\ \varepsilon_{210} \end{bmatrix}$$

Then Y=Xβ+E.

Example 1 (Cont.): A design engineer is interested in determining any difference in the mean of burning times of chemical flares of two different formulations. She found the burning times of 10 flares for each of the two formulations. The results are given in the following table:

Type1	Type2
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59	74
66	59
75	82
82	65
70	79

```
> Typ1 = c(65, 82, 81, 67, 57, 59, 66, 75, 82, 70)
> Typ2 = c(64, 56, 71, 69, 83, 74, 59, 82, 65, 79)
> Y = c(Typ1,Typ2)
> f = factor(rep(c("Type1","Typ2"), each=10))
> model = lm(Y~f)
> model.matrix(model)
> > anova(model)
Analysis of Variance Table

Response: Y
          Df   Sum Sq   Mean Sq   F value   Pr(>F)
f          1    0.2     0.200     0.0023    0.9622
Residuals 18  1562.0    86.778
```

What Have We Learned?

An **experiment** is a scientific procedure undertaken to make a discovery or to determine something.

The measurable outcome of the experiment is called **response**.

The variables whose effect on the outcome is to be studied is called **factor**.

Different values of a factor is a called **levels** of the factor.

A **treatment** is combinations of factor levels

What Have We Learned? (Cont.)

The three basich principles of experimental design are:

- **Replication:** A repetition of the basic experiment is called replication.
- **Randomization:** Both the allocation of the experimental material and the order in which the individual runs or trials of the experiment are to be performed are randomly determined.
- **Blocking:** Blocking is a technique used to increase the precision of an experiment. A block is a portion of the experimental material that should be more homogeneous than the entire set of materials. Blocking involves making coparisons among the condition of interest in the experiment within each block.